Mustafa Mehmood

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Experience

Applied Research Scientist — *London, UK*
Thomson Reuters — *June 2025 – Present*

- Building agentic systems for CoCounsel, TR's AI legal assistant, that extract risks across M&A due diligence documents, working closely with lawyers to encode their methodologies, identifying 90%+ of material risks.
- Developed a framework-agnostic skill library that breaks due diligence into composable, reusable sub-skills for risk analysis, fact extraction, and document classification, orchestrated as sub-agents within an agent harness and validated through SME evaluations.
- Engineered reusable agent tools exposed over MCP for document retrieval, citation verification, cross-reference resolution, and redline comparison, shared across CoCounsel agents.
- Devised context-engineering strategies for documents exceeding context limits: retrieval-then-reason, progressive summarization, and state passed between sub-agents.
- Designed an adversarial evaluation pipeline for agentic solutions with 90%+ failure detection rate (0.78 Cohen's Kappa vs. expert grades), replacing manual grading and reducing evaluation cycles from days to hours.
- Fine-tuned LLMs for legal use cases including long-form contract drafting, customization, and summarization.

LLM Data Integrity Specialist — *Nottingham, UK*
Outlier AI — *May 2024 – May 2025*

- Improved tool use in frontier models through RLHF training pipelines, enabling them to chain and execute multi-step actions from high-level user intent, supporting AI development at firms including Google and Meta.
- Curated instruction tuning and preference datasets to drive a 20% improvement in instruction adherence.

Data Scientist Intern (Part-Time) — *Selangor, Malaysia*
Department of Neuroscience, University of Nottingham Malaysia — *June 2021 - Aug 2023*

- Engineered feature extraction pipelines to transform raw EEG and eye-tracking clinical data into structured features: coherence, Hjorth parameters, and spectral power bands for EEG; gaze patterns and fixation analysis for eye-tracking.
- Built EEG-based diagnostic classifiers for neurological and mental health disorders, reaching 80% accuracy for Parkinson's and 82% for Schizophrenia.
- For autism screening, represented gaze scanpaths as images and trained a custom transfer-learning computer vision model, benchmarked against a PCA-based classical pipeline, to reach 92% accuracy.

Education

University of Nottingham — *Nottingham, UK*
MSc** Artificial Intelligence - **Grade: Distinction — *Sept 2023 – Dec 2024*
BSc** Computer Science **(Hons) — *Sept 2020 – Aug 2023*
Key Modules: Machine Learning, Computer Vision, Natural Language Processing, Big Data, Information Visualisation

Key Projects

Sherlock-AI: Character Emulation with LLMs — Website Repo

- Designed a resource-efficient, transferable character emulation pipeline: fine-tuned a quantized LLaMA model with *LoRA* for Sherlock Holmes emulation, automating dialogue extraction and synthetic data generation from public domain texts.
- Implemented ReAct-style reasoning for multi-step deductive analysis with dual inference support (OpenRouter API and local GGUF), optimizing for performance and cost across cloud and local setups.
- Built an interactive story mode where users play as characters in branching Sherlock Holmes mysteries through structured decisions and free-form actions, using templated prompts with streaming structured output to deliver real-time scene generation, configurable writing styles, and auto-generated context summaries.
- In A/B testing, 72.4% of users favoured the fine-tuned model over baseline for authenticity, with perfect usability scores.

CourseCompanion: RAG-Powered Course Assistant — Repo

- Developed an educational platform enabling professors to create AI teaching assistants that auto-generate knowledge bases from course materials, featuring chat interfaces, FAQ system, and management dashboards; built with Flask and PostgreSQL using FAISS-backed RAG with LangChain and Flair.

DeepGaze: Autism Diagnosis with ML and Webcam Eye-Tracking — Demo Video Repo

- Built a modular Flask web app for webcam-based eye-tracking data collection, visualization, real-time prediction, and model retraining, operationalizing the autism-detection method from my publication.

Publication

Mehmood, M., Amin, H. U. (2024). Pre-diagnosis for Autism Spectrum Disorder Using Eye-Tracking and Machine Learning Techniques. *Advances in Brain Inspired Cognitive Systems*, Springer Nature Singapore.

Technical Skills

Programming Languages: Python, C, Java, JavaScript, SQL, HTML/CSS

Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, trl), scikit-learn, LangChain, LangGraph, Pydantic AI, spaCy, vLLM, FastAPI, Flask, pandas, NumPy, FAISS, Weights & Biases, Gradio/Streamlit, Claude Agent SDK

Tools & Platforms: Git/GitLab CI, Docker, AWS (SageMaker, EC2, Lambda, Bedrock), Spark/Databricks, PostgreSQL, Pinecone, Runpod, DigitalOcean, MCP, Agent Harness Design, Claude Code, Cursor